Republic of the Philippines
Department of Environment and Natural Resources
MINES AND GEOSCIENCES BUREAU
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FLOOD THREAT ADVISORY

To: Brgy. Chairperson
Brgy. Tumana
City of Marikina
Metropolitan Manila

Date: October 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Barangay Hall and vicinity	14° 39' 43.3" N, 121° 05' 56.4" E	High	The Ampalaya Creek bounds the barangay along its eastern and southern margin. Multiple observation points in its periphery share similar flooding situations during Ondoy and Ulysses. During the former, floodwaters reached and exceeded the second floor of two-storey houses, which subside the following morning. The barangay official noted that the river water level was higher during Ulysses, but the flood heights reported within the barangay was lower, although still exceeded 2m. This reduction in flood height can be attributed to the installation / improvement of flood mitigation structures along Marikina River.
Ampalaya Creek (Angel Santos St.)	14° 39' 13.8" N, 121° 06' 02.1" E	Very High	
Confluence of Amapalaya Creek and Marikina River	14° 39' 21.8" N, 121° 05' 40.9" E	Very High	The vicinity of the confluence of Ampalaya Creek and Marikina River is characterized by flood heights exceeding 2m during both Ondoy and Ulysses. In the more recent event, the installed dike was reportedly breached by overflowing river water, which inundated motor vehicles that were atop said dike as a precaution.
Farmers II	14° 39' 24.2" N, 121° 05' 48.7" E	Very High	In Farmers II, during Ulysses, flood heights exceeded 2m and breached the second floor of the house.
Block 53-54 (Pipino St.)	14° 39' 32.0" N, 121° 05' 49.7" E	Very High	Block 53-54 / Pipino St. is a relatively depressed compared to the main Pipino St. and is more proximal to Marikina River. Flood heights during Ondoy reached the flooring of the terrace of the house in picture. Households along the more elevated Pipino St. still experienced flood heights greater than 2m during Ulysses. At the time of assessment, ongoing construction in a section of Marikina River was noted).

Remarks / Recommendations:

- ✓ It is highly recommended to implement appropriate easements in construction of settlements or other infrastructure along rivers/creeks as specified in the Presidential Decree No. 1067 known as Water Code of the Philippines.
- ✓ Improvement and proper maintenance of existing drainage system in the barangay are recommended to prevent inundation of residential areas, critical facilities, and other assets. Avoid improper waste disposal, particularly along drainage canals as it may cause clogging and obstruction of water flow. Ensure regular clearing of natural and artificial drainage pathways, or prior to an impending tropical cyclone or other rain-bearing weather systems.
- ✓ Construction of appropriate and well-studied engineering measures (e.g., dike, flood wall, etc.) should be implemented to minimize the rate of erosion along rivers/creeks. Ensure that the structure follows engineering standards and has structural integrity. Also, damages in the structural mitigation measures should be reported to concerned agency/office responsible for the installation. The LGU is also encouraged to maintain such measures or reinforce identified damages, with the strong support of the barangay officials and active engagement of the communities and the private sector.
- ✓ Install a warning signage alerting motorists and residents of the danger of crossing bridges susceptible to flash flood during heavy rainfall events.
- ✓ The CDRRMO, together with the BDRRMO, should conduct Information, Education, and Communication (IEC) Campaign on flood hazards to flood-prone communities, particularly the households residing proximal to the river and irrigation canals.
- ✓ Strengthen and improve flood warning system with corresponding warning levels for pre-emptive evacuation purposes. The barangay should have constant communication with nearby and/or upland/upstream barangays during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas.
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of tropical cyclones and other prevailing weather disturbances. This may include City Disaster Risk Reduction Management Office (CDRRMO), Office of the Civil Defense (OCD) NCR, and other related agencies.

Prepared by:

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Received by:
Designation:
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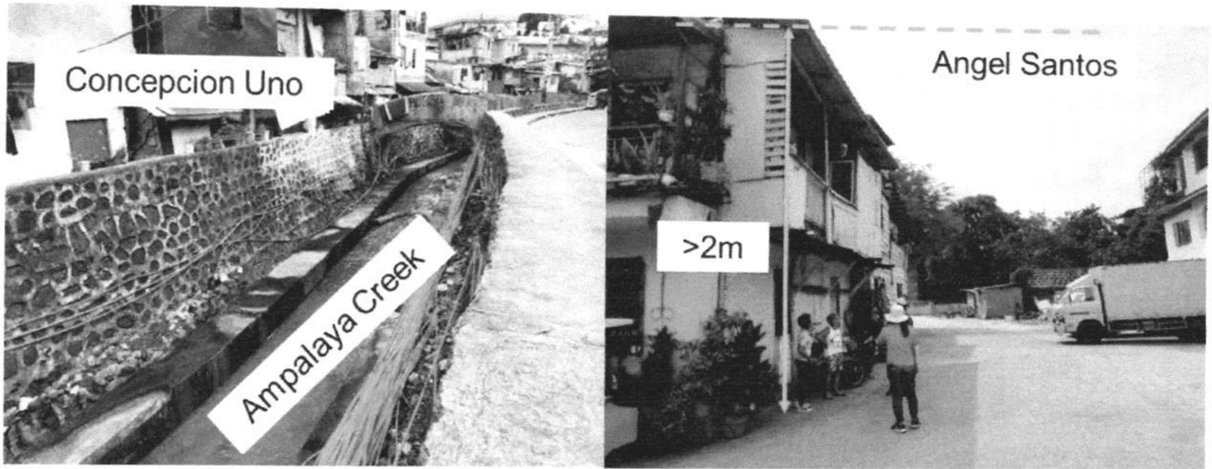
BARANGAY TUMANA



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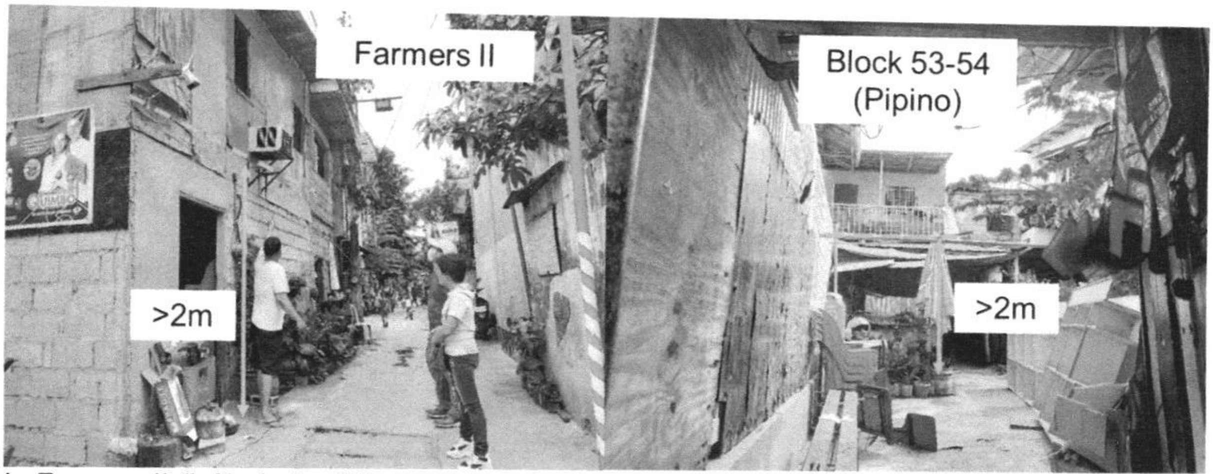
The vicinity of the confluence of Ampalaya Creek and Marikina River is characterized by flood heights exceeding 2m during both Ondoy and Ulysses. In the more recent event, the installed dike was reportedly breached by overflowing river water, which inundated motor vehicles that were atop said dike as a precaution (GPS Coordinates: 14° 39' 21.8" N, 121° 05' 40.9" E).



In Farmers II *(left)*, during Ulysses, flood heights exceeded 2m and breached the second floor of the house in picture (GPS Coordinates: 14° 39' 24.2" N, 121° 05' 48.7" E). While Block 53-54 / Pipino St. *(right)* is a relatively depressed compared to the main Pipino St. and is more proximal to Marikina River. Flood heights during Ondoy reached the flooring of the terrace of the house in picture. Households along the more elevated Pipino St. still experienced flood heights greater than 2m during Ulysses. At the time of assessment, ongoing construction in a section of Marikina River was noted (GPS Coordinates: 14° 39' 32.0" N, 121° 05' 49.7" E).

Recommendations:

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